- *Jobs*—The background tasks that implement your logic.
- *Triggers*—Control *when* a job runs based on a schedule, such as "every five minutes" or "every second Tuesday." A job can have multiple triggers.
- *Job factory*—Responsible for creating instances of your jobs. Quartz.NET integrates with ASP.NET Core's DI container, so you can use DI in your job classes.
- *Scheduler*—Keeps track of the triggers in your application, creates jobs using the job factory, and runs your jobs. The scheduler typically runs as an `IHostedService` for the lifetime of your app.

Background services vs. cron jobs

It's common to use cron jobs to run tasks on a schedule in Linux, and Windows has similar functionality with Task Scheduler, used to periodically run an application or script file, which is typically a short-lived task.

By contrast, .NET apps using background services are designed to be long-lived, even if they are used only to run tasks on a schedule. This allows your application to do things like adjust its schedule as required or perform optimizations. In addition, being long-lived means your app doesn't only have to run tasks on a schedule. It can respond to ad hoc events, such as events in a message queue.

Of course, if you don't need those capabilities and would rather not have a long-running application, you can use .NET in combination with cron jobs. You could create a simple .NET console app that runs your task and then shuts down, and you could schedule it to execute periodically as a cron job. The choice is yours!

In this section I show you how to install Quartz.NET and configure a background service to run on a schedule. Then I explain how to enable